

YORKTON, SK, Canada

WMO: 712920

Lat: 51.2601N

Lon: 102.4615W

Elev: 498

StdP: 95.48

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.5	-30.0	-37.3	0.1	-32.4	-34.7	0.2	-29.8	12.9	-9.4	11.4	-11.5	3.3	300	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	29.3	19.5	27.4	18.6	25.8	17.5	21.2	26.9	19.9	25.6	18.8	24.0	5.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.8	24.3	17.9	13.6	22.9	16.8	12.7	21.4	64.2	26.9	59.4	25.5	55.4	23.9	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.7	8.7	DB	-36.5	32.8	3.1	1.6	-38.7	34.0	-40.5	34.9	-42.3	35.8	-44.5	37.0	(4)
(5)			WB	-36.6	23.7	3.1	1.7	-38.8	24.9	-40.6	25.9	-42.3	26.9	-44.5	28.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-15.8	-13.9	-7.1	2.9	10.3	15.7	18.4	17.3	12.3	4.2	-5.4	-13.1	(6)	
	DBStd	13.69	8.65	7.77	7.78	6.17	4.84	3.59	3.04	3.42	4.46	5.05	6.89	7.92	(7)	
	HDD10.0	3674	800	670	529	223	56	3	0	1	27	188	463	715	(8)	
	HDD18.3	5964	1058	903	787	464	251	93	37	60	187	437	713	973	(9)	
	CDD10.0	841	0	0	0	8	66	174	262	227	94	9	0	0	(10)	
	CDD18.3	89	0	0	0	0	2	14	40	28	4	0	0	0	(11)	
	CDH23.3	1039	0	0	0	3	61	164	387	331	91	3	0	0	(12)	
	CDH26.7	239	0	0	0	0	11	32	87	91	18	0	0	0	(13)	
(14) Wind	WSAvg	4.3	4.4	4.3	4.4	4.6	4.7	4.1	3.7	3.7	4.1	4.5	4.4	4.2	(14)	
(15) Precipitation	PrecAvg	457	23	17	28	23	53	70	64	58	50	26	20	23	(15)	
	PrecMax	593	55	48	58	82	115	152	141	141	109	77	54	41	(16)	
	PrecMin	272	1	4	4	0	4	11	11	14	2	2	0	7	(17)	
	PrecStd	86	16	11	13	16	28	39	36	39	29	20	13	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	3.8	11.4	22.4	28.1	29.9	31.4	32.1	29.1	22.3	12.2	4.7	(19)	
		MCWB	1.7	1.7	6.4	12.0	15.4	20.4	21.5	20.0	18.3	13.6	7.4	1.9	(20)	
	2%	DB	1.4	1.2	7.2	18.7	25.0	27.2	29.0	29.2	26.1	17.9	8.0	2.1	(21)	
		MCWB	-0.1	-0.1	4.0	10.1	13.9	17.9	20.6	19.4	16.2	10.5	4.4	0.4	(22)	
	5%	DB	-0.9	-0.7	4.5	15.7	22.2	25.1	27.2	26.9	23.2	14.7	5.2	-0.3	(23)	
		MCWB	-2.0	-1.7	2.3	8.2	12.6	16.7	19.3	18.5	15.2	9.2	2.6	-1.6	(24)	
	10%	DB	-3.3	-2.7	2.5	12.6	19.8	23.2	25.5	24.8	20.6	12.0	3.0	-2.5	(25)	
		MCWB	-4.1	-3.5	0.8	6.8	11.3	15.7	18.5	17.4	13.8	7.9	1.4	-3.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	1.8	7.0	12.6	16.9	22.5	23.3	22.1	19.5	14.1	7.8	2.1	(27)	
		MCDB	3.4	3.1	10.8	21.1	23.7	27.3	28.4	28.3	26.3	20.6	11.7	4.5	(28)	
	2%	WB	-0.1	0.1	4.1	10.5	15.0	19.4	21.7	20.6	17.6	11.6	4.8	0.5	(29)	
		MCDB	1.4	1.1	7.0	17.3	22.4	24.7	27.1	27.0	23.7	16.3	7.4	2.1	(30)	
	5%	WB	-2.0	-1.7	2.4	8.9	13.5	17.8	20.4	19.4	16.0	9.7	2.9	-1.5	(31)	
		MCDB	-0.9	-0.7	4.4	14.6	20.8	23.0	25.7	25.2	21.8	14.1	4.7	-0.4	(32)	
	10%	WB	-4.1	-3.5	0.9	7.3	12.1	16.6	19.2	18.2	14.3	8.0	1.3	-3.3	(33)	
		MCDB	-3.2	-2.7	2.4	12.3	18.3	21.5	24.2	23.5	19.7	11.7	3.0	-2.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.3	9.5	11.0	12.6	10.9	11.3	12.2	11.9	9.9	8.1	8.2	(35)	
		MCDBR	9.8	9.6	10.3	15.9	16.4	14.2	13.9	15.6	16.5	15.3	11.2	9.9	(36)	
	5% WB	MCWBR	8.9	8.4	7.7	8.9	7.7	6.8	7.1	7.5	8.6	9.1	8.0	8.6	(37)	
		MCWBR	9.9	9.0	10.0	14.7	14.5	12.1	12.7	13.7	15.0	14.5	10.4	9.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.246	0.238	0.281	0.318	0.325	0.348	0.364	0.370	0.328	0.280	0.248	0.234	(40)	
		taud	2.288	2.352	2.375	2.435	2.469	2.431	2.400	2.369	2.469	2.565	2.485	2.346	(41)	
	Ebn at Noon		791	918	928	926	930	907	884	859	865	860	804	757	(42)	
		Edh at Noon	64	79	95	102	104	109	111	108	87	65	53	51	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.25	2.36	3.78	4.94	5.79	5.92	6.27	5.08	3.76	2.19	1.15	0.88	(44)	
	RadStd		0.10	0.20	0.31	0.35	0.51	0.47	0.35	0.42	0.33	0.29	0.15	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.54	N/A	N/A	N/A	N/A

Nomenclature: See separate page